

THE EVACUATION RISK AND COST FRAMEWORK

ESTIMATING THE HUMAN AND FINANCIAL COST OF CIVILIAN EVACUATION IN ARMED CONFLICT

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A humanitarian organisation facing a besieged city has to answer a question that no manual answers well: what would it actually take to move these people, and what would it cost to keep them alive if they cannot be moved? Both questions have to be answered before a funding appeal can be written, and both are usually answered from memory, from a previous operation in a different country, or from an experienced coordinator's intuition. ERCF is a research prototype that attempts to put numbers on both and to show its working.

KEY FIGURES

7

weighted dimensions a planner scores from 1 to 5, with kinetic threat carrying a quarter of the total

31

documented historical conflicts in the corpus, sixteen of which were used to fit the mortality model

0.855

R-squared in logarithmic space, against only 44 per cent of fitted cases landing within a factor of two

90

days of planning window, beyond which the model sits outside the range of its own calibration data

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EXECUTIVE SUMMARY

ERCF is a research prototype: experimental software, not an operational system. It helps a humanitarian planner estimate two things side by side. The first is the one-time cost of organising an evacuation. The second is the daily cost of keeping people supplied if they stay. It then compares the two.

The planner describes the situation using seven factors, called dimensions and labelled D1 to D7. Each is scored from 1 to 5. They cover the intensity of fighting, the mobility of the population, whether the armed parties have authorised movement, the state of roads and logistics, the safety of the destination, how fast the window is closing, and how good the available information is. These seven scores drive everything else the tool produces.

The seven scores are combined into a single weighted number, which is placed in one of five risk levels from Level 0 to Level 4. That level then selects the cost rates, loss rates and mortality rates used in the rest of the calculation. A planner who moves one slider sees the entire cost estimate change.

The distinctive output is the break-even analysis. Evacuation is a large one-time expense. Assistance in place is a smaller expense repeated every day. Somewhere in the future, if the crisis persists, the cumulative daily

cost overtakes the one-time cost. ERCF calculates the day on which that happens, which is a question funding appeals have to answer and rarely answer with arithmetic.

ERCF also produces an estimate of deaths and injuries among a population that does not leave. This estimate was fitted against sixteen documented historical conflicts. Its author is explicit that this part of the tool is indicative and that the financial estimates are considerably more reliable than the mortality ones.

The project takes an explicit ethical position, stated in its own documentation and repeated in the interface. It estimates the cost of logistics. It does not place a monetary value on human life, and it does not tell a user whether an evacuation is worth carrying out. The labels it produces are descriptive, not instructions.

The software is substantially built rather than aspirational. The repository contains roughly eleven thousand lines of working Python, a browser interface, a reproducible calibration pipeline, a database of thirty-one documented historical conflicts, and connections to four external data services. It runs as a live public demonstration. What remains incomplete is documented openly in the project's own backlog.

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BACKGROUND AND RATIONALE

An organisation deciding whether to support an evacuation needs three figures quickly: what evacuation would cost, what staying would cost, and where those two lines cross. Each figure exists somewhere. Sphere standards give water and food quantities per person. The World Food Programme publishes what its air service costs per passenger kilometre. Case studies record what particular operations spent. But these sit in separate documents written for separate purposes, and nobody has assembled them into a single calculation that a planner can run in an afternoon.

The project's concept document states the position directly: these estimates exist in operational manuals and case studies, but no publicly available tool integrates them with a historical calibration dataset and a framework grounded in International Humanitarian Law. ERCF is an attempt to build that integration and to publish the assumptions rather than hide them.

One finding recorded in the source code is worth restating for a policy audience. When the author searched for a published figure describing how much more expensive it is to deliver humanitarian assistance inside an active conflict than in a stable setting, no such figure was found. The International Federation of Red Cross and Red Crescent Societies confirms in a 2021 methodology paper that the per-capita cost of response is statistically significantly higher in conflict, but publishes no ratio. The World Food Programme publishes per-tonne logistics costs that imply a premium of roughly 1.23 times for Sudan in 2023 against an Eastern Africa regional average. Security costs, which the model treats as the larger component, appear only in fragments: an armed escort in Yemen at up to 3,000 US dollars per convoy on one route, security spending in Somalia in 2017 estimated at around 400 million US dollars per year against total humanitarian operations of 700 to 900 million. The code notes that a single published multiplier for active-conflict delivery does not appear to exist, and that this model may be among the first attempts to quantify it explicitly. That is an honest statement of a weakness and of a contribution at the same time.

ERCF builds the calculation anyway, using the best documented values available, and tags every single number with the strength of the evidence behind it. A planner reading the interface can see which figures are validated against a published standard, which are estimated from indirect evidence, and which are unvalidated assumptions the author could not support from any public source. That tagging is the tool's principal safeguard against being trusted more than it deserves.

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OBJECTIVES

The tool is designed to:

- Produce a rapid, itemised cost estimate for a civilian evacuation operation, broken into transport, fuel, personnel, food, water, shelter, medical supplies, communications and contingency.
- Produce a parallel estimate for the cost of assisting a population that remains in the conflict zone, including the extra cost that conflict access conditions impose and the proportion of supplies that never arrives.

- Identify the break-even point at which continuing in-zone assistance becomes more expensive than a one-time evacuation, so that funding appeals can be argued from arithmetic rather than assertion.
- Place the current situation against documented historical operations, so that a planner can see which past conflicts a scenario most resembles.
- Make every parameter, source and confidence level visible in the interface rather than buried in the code.
- Keep the estimate of financial cost separate from the estimate of mortality, so that a funding calculation is never confused with a casualty prediction.

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HOW ERCF WORKS: THE SEVEN DIMENSIONS

The whole tool rests on seven scores. Each is a number from 1 to 5 that the planner sets by moving a slider. The dimensions were chosen, according to the concept document, to capture the factors that field coordinators consistently cite when assessing whether an evacuation is feasible, and deliberately to require no real-time data feed or classified intelligence. A planner with a situation report and professional judgment can set all seven.

D1, Kinetic Threat. How much direct violence the civilian population is exposed to. A score of 5 means active, sustained attack. This is the dimension that drives almost everything downstream: it determines whether movement is physically survivable at all.

D2, Mobility Constraints. Also called Vulnerability. How much of the population cannot move unaided. A high score means many people who are elderly, very young, disabled, chronically ill or otherwise dependent on assisted transport. This dimension is what forces the operation to allocate medical buses and ambulances rather than ordinary buses.

D3, Authorization. Whether the armed parties have consented to civilian movement. The scale runs the same direction as the others: a high D3 score means authorization is a serious problem, that is, that consent is absent or unreliable. Without consent, an evacuation is both unlawful and practically blocked.

D4, Logistics. The state of roads, bridges, vehicles, fuel supply and the supporting infrastructure. The project's documentation observes that

logistics collapse, not danger as such, has empirically been the leading cause of delayed evacuations, citing Mosul in 2016 and Aleppo in 2016.

D5, Destination. Whether the place people would be moved to is genuinely safe and able to receive them. The reason this dimension exists at all is Srebrenica in 1995, cited in the code as the canonical case in which evacuation to a nominally safe area became the site of a massacre. Moving civilians into danger is a distinct harm, not a lesser version of leaving them where they are.

D6, Urgency. How fast the window for organised movement is closing.

D7, Information. How poor the information environment is. A high score means communications failure, rumour, and an inability to coordinate.

The seven scores are combined into a single number by multiplying each by a weight and adding the results. The weights add up to 1.00, so the resulting score stays on the same 1 to 5 scale as the inputs.

DIMENSION	FACTOR	WEIGHT
D1	Kinetic Threat	0.25
D2	Mobility Constraints	0.15
D3	Authorization	0.15
D4	Logistics	0.15
D5	Destination	0.15
D6	Urgency	0.10
D7	Information	0.05

$$\text{ERCF} = (D1 \times 0.25) + (D2 \times 0.15) + (D3 \times 0.15) + (D4 \times 0.15) + (D5 \times 0.15) + (D6 \times 0.10) + (D7 \times 0.05)$$

The seven weights sum to 1.00.

D1 carries the largest weight, one quarter of the total, because direct physical threat is treated as the primary driver of evacuation necessity, and because it is the dimension that alone determines whether movement is safe. It is tied to the precautionary obligations in Articles 57 and 58 of Additional Protocol I.

D2, D3, D4 and D5 each carry fifteen per cent and form an equal group. The argument is that mobility, consent, logistics and destination safety are each individually capable of making an evacuation impossible, so none should dominate the others. D3 is explicitly capped at fifteen per cent rather than being set higher, because the most extreme authorization failures are already captured elsewhere in the tool by a hard trigger.

D6 carries ten per cent, deliberately below the equal group. The reasoning is that urgency is largely absorbed by D1 in the most extreme situations, and that the tool captures urgency more sharply through the hard trigger and through a separate extraction-probability floor than a linear weight would. D7 carries five per cent, the lowest. Poor information raises coordination cost and panic risk, but on its own it does not determine whether an evacuation is necessary or possible.

An important caveat is stated by the author in the code itself and reproduced in the interface as a red marker beside every weight: all seven weights are modelled estimates. No published framework in International Humanitarian Law or humanitarian operations specifies numerical weights for these factors, so there was nothing to copy. The author's own backlog lists expert-panel validation of these weights as outstanding research work. A reader should treat the weights as a reasoned starting proposal, not as an established standard.

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FROM SEVEN SCORES TO A RISK LEVEL

The weighted score is placed into one of five bands, each with a label and a NATO-doctrine equivalent used in military planning.

LEVEL	LABEL	WEIGHTED SCORE	NATO EQUIVALENT
Level 0	Baseline and Monitoring	1.5 or below	permissive and stable
Level 1	Low Risk and Advisory	above 1.5 up to 2.5	permissive but degrading
Level 2	Moderate Risk and Watchful	above 2.5 up to 3.5	uncertain
Level 3	High Risk and Contested	above 3.5 up to 4.2	hostile, partial
Level 4	Critical and Emergency	above 4.2	hostile, imminent

There is one override. If D1 and D6 are both at 4.5 or above, that is, if violence is extreme and the window is closing at the same time, the score is floored at 4.21, which forces Level 4 regardless of how favourable the other five dimensions look. This exists because the linear weighted average would otherwise allow good logistics and a safe destination to pull an imminent massacre down into Level 3.

The level is not merely a label. It selects the numerical rates used everywhere downstream: the security escort ratio, the daily cost of assistance per person, the access multiplier, the supply loss rate, the injury rate, the mortality base rate, and the probability of emergency extraction. Almost every figure the tool produces changes when the level changes. This is a design choice worth noticing, because it means the five-band classification carries a great deal of weight and the boundaries between bands, at 1.5, 2.5, 3.5 and 4.2, are themselves modelled judgments rather than empirical findings.

Alongside the composite score, the tool computes three sub-indexes that keep distinct questions apart. This was added on the recommendation of the project's academic reviewers. Risk Severity asks how dangerous the situation is for civilians, and is built from D1, D2 and D6 only, rescaled back onto a 1 to 5 range. Feasibility asks whether people can realistically move, and is built from D3, D4 and D5, but inverted: because a high D3, D4 or D5 score means bad conditions, each is subtracted from 6 before being used, so that a high feasibility number means a genuinely open corridor. Information Quality

is simply 6 minus D7, so that a high number means good situational awareness.

Keeping severity and feasibility apart matters. A situation can be extremely dangerous and simultaneously impossible to evacuate. Blending the two into one score would make that case look moderate, which is precisely the case that should escalate most urgently. The tool instead places the scenario in a four-cell matrix. High severity with high feasibility returns evacuate immediately. High severity with low feasibility returns shelter in place and negotiate a corridor urgently, with the reasoning that forced movement without a safe corridor risks greater harm than staying, and with a note that the precautionary obligations under Articles 57 and 58 of Additional Protocol I apply while negotiation continues. Low severity with high feasibility returns facilitate voluntary departure, and explicitly says do not mandate evacuation. Low severity with low feasibility returns monitor and reassess daily.

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THE COST OF EVACUATING: EVERY VARIABLE EXPLAINED

This is the part of the tool with the strongest evidentiary foundation. The planner supplies four things: the population at risk, the percentage of that population who are vulnerable, the distance in kilometres to safety, and the terrain quality. Everything else is derived.

The population is split into vulnerable and non-vulnerable. Non-vulnerable people are assigned to standard buses at 50 people per bus. Vulnerable people are assigned to medical buses at 20 people per bus, and ambulances at one per 150 vulnerable people. The bus capacities are marked in the code as operational assumptions not validated against field data. The ambulance ratio has a history worth recording: it was originally set at one ambulance per 40 vulnerable people, and was revised to one per 150 after the author found that no published field standard exists at all and that the original figure was three to five times above what documented practice, in particular a study of the Kosovo operation, actually showed.

The number of medical buses and ambulances is then multiplied by a factor derived from D2, the mobility dimension. At D2 of 1 the factor is 0.8, at 2 it is 1.0, at 3 it is 1.3, at 4 it is 1.8, and at 5 it is 2.5. In plain terms, a population with severe mobility constraints needs roughly two and a

half times the assisted transport of a baseline population. This factor is described in the code as estimated with no primary source.

Staffing is set by ratio, and the security ratio tightens sharply as danger rises. The medical staffing figure is the Sphere Handbook 2018 standard for clinical officers in emergency settings, and it was corrected upward during development: an earlier version used one per 500, half the Sphere standard, with no documented justification.

ROLE	ALLOCATION	DAILY RATE, US DOLLARS
Security	none at Level 0, one per 500 civilians at Level 1, per 200 at Level 2, per 100 at Level 3, per 50 at Level 4	300
Medical staff	one per 250 people	200
Paramedics	one per 100 people	150
Drivers	one per vehicle	50

Daily rates are total cost to the operation, not take-home pay. All four are tagged as estimated.

The code carries an unusually candid note about the rates: every rate implicitly assumes national staff or a lower-cost international non-governmental deployment at roughly the level of Médecins Sans Frontières, and full United Nations international professional staff, once daily subsistence allowance and danger pay are included, would cost three to six times more. The author records that this assumption is not stated in the interface and should be flagged.

Water is calculated at 20 litres per person per day for three days. This is the UNHCR full planning standard, chosen deliberately over the Sphere emergency minimum of 7.5 to 15 litres because an evacuation is a planned operation rather than a first-response emergency. Food is calculated at 0.45 kilograms of dry food per person per day for three days, which is the dry-weight equivalent of the Sphere minimum of 2,100 kilocalories per person per day. Tents are provided at one per five people, which at the Sphere standard of 3.5 square metres per person gives 17.5 square metres per tent and is internally consistent. Basic medical kits are provided at one per 100 people, and trauma kits at one per 50 people when the risk level is 3 or

above, or one per 200 otherwise. Radios are provided at one per five vehicles plus a fixed five for coordination.

The unit costs applied to those quantities are set out below, each with the evidence the project records behind it. Fuel is the one line with a consumption figure of its own: it is calculated at 0.35 litres per kilometre per vehicle for a return journey before the per-litre price is applied.

ITEM	UNIT COST	EVIDENCE RECORDED IN THE CODE
Standard bus	200 US dollars each	
Medical bus	400 US dollars each	Revised upward after research found no primary source for the earlier value; sits at the lower bound of documented field ranges for medically-equipped vehicles.
Ambulance	700 US dollars each	Revised upward on the same basis as the medical bus.
Fuel	1.20 US dollars per litre	Revised down from 1.50 on the basis of ACAPS reporting of Yemeni consumer fuel prices in 2022.
Food	3 US dollars per kilogram	Flagged by the project's own parameter registry as having no citation behind it, even though the quantity does.
Water	0.05 US dollars per litre	Field evidence gathered in 2026 found real water trucking costs of 2 to 23 US dollars per cubic metre, all below the model's baseline; the lower figures were not adopted pending further validation.
Tent	380 US dollars each	Raised from an earlier 150 once that figure was found to describe a tarpaulin kit suitable for a week rather than a standard tent; 380 sits just below the 400 dollar replacement cost UNHCR gave publicly in 2022.
Basic medical kit	21 US dollars	Derived from the WHO and UNICEF Interagency Emergency Health Kit costing, revised down from an earlier 50 that was roughly seven times too high for a three-day convoy.
Trauma kit	200 US dollars	Unvalidated. The ICRC does not publish per-kit pricing.

ITEM	UNIT COST	EVIDENCE RECORDED IN THE CODE
Radio	500 US dollars each	Within the documented procurement range for professional handheld VHF units.

Three multipliers are applied on top. Terrain multiplies transport and fuel costs, with five levels running from 4.0 for the worst terrain down to 1.0 for the best, and 2.5, 1.7 and 1.2 in between. The lower end of this range is consistent with published road-condition cost models; the upper end of 4.0 is an expert estimate consistent with a World Food Programme figure showing per-tonne delivery costs in the Central African Republic, South Sudan and the Democratic Republic of the Congo running about five times a standard country average, though that figure mixes terrain, access and security together.

Season adjusts terrain further. If the operation starts in a month the tool classifies as a closure period for that latitude, the terrain multiplier is boosted by 50 per cent for the worst terrain, 30 per cent for the second worst and 20 per cent for middling terrain. Closure periods are set by latitude: December to March above 30 degrees north, June to September below 30 degrees south, and April to October in the tropics for wet season. The author describes these as broad regional approximations. The most useful output here is not the cost boost but a flag: the worst terrain in a closure period is marked potentially impassable, which is an operational warning rather than a number.

D4, logistics, adds 10 per cent to transport and fuel for each point above 1, so a D4 of 5 adds 40 per cent. D5, destination, changes the number of tents needed: a destination with good existing infrastructure halves the tent requirement, a destination with none doubles it. Both are marked estimated with no primary source. Finally, 15 per cent is added to the whole subtotal as contingency, which the code justifies as the lower bound of the 15 to 20 per cent standard used for high-risk projects.

A worked example is carried in the code. For 10,000 people, 20 per cent of them vulnerable, at Level 2, moving 50 kilometres: 160 standard buses, 100 medical buses and 50 ambulances, 310 vehicles in total; 50 security staff, 20 medical staff, 100 paramedics and 310 drivers; 10,850 litres of fuel, 15,000 kilograms of food, 450,000 litres of water, 2,000 tents and 67 radios. Transport comes to 92,500 US dollars, fuel to 16,275, personnel to 34,000, food to 45,000, water to 22,500, shelter to 300,000, medical to

15,000 and communications to 33,500, giving a subtotal of 558,775, a contingency of 83,816, and a total of roughly 643,000 US dollars, or about 64 dollars per person. Shelter dominates, which is a useful thing for a planner to be able to see at a glance.

07

THE COST OF STAYING

The second calculation estimates what it costs an organisation to keep a population alive where it is. It has three parts.

The first is supply delivery. The starting point is 3.50 US dollars per person per day, a multi-sector figure covering food, water, health, shelter and coordination. The World Food Programme’s own global average for food and cash assistance alone was 42 US cents per beneficiary per day in 2023, and the code explains the difference between the two figures at some length: at the full Sphere water standard, water alone dominates supply weight, and applying the World Food Programme’s per-tonne Sudan delivery cost to that weight gives 3.28 US dollars per person per day, within six per cent of the model’s figure. That is a genuinely careful piece of reasoning.

The baseline is then multiplied by an access multiplier that represents how much more expensive delivery becomes as conflict intensifies. The second part of the calculation, supplies that never arrive, is then added as a loss rate representing goods destroyed, looted or simply undelivered. Both ladders are selected by the risk level.

RISK LEVEL	ACCESS MULTIPLIER	SUPPLY LOSS RATE
Level 0	1.0	5 per cent
Level 1	1.5	5 per cent
Level 2	2.0	15 per cent
Level 3	3.6	30 per cent
Level 4	4.0	50 per cent

The access multipliers were revised downward during development, from earlier values of 3.0, 5.0 and 8.0 for the top three levels, after the author's research established that documented sources support only about 2.5 to 3.6 times at Level 3 and nothing above about 4 times at Level 4. Level 4 remains marked as directionally plausible but unconfirmed, and the project's backlog lists it as the outstanding unvalidated cost parameter. A further penalty is applied from D4, and the terrain and seasonal multipliers apply here too, because bad roads make delivery expensive whether people are leaving or staying.

Only the lowest of the loss rates has any published anchor: OCHA monitoring of Gaza during the ceasefire period in late 2025 recorded under two per cent of cargo looted or intercepted under active monitoring, and the model uses 5 per cent as a planning buffer including spoilage. The three higher figures are internal planning estimates with no published equivalent that the author could find. A low D3, meaning consent is absent, adds further to the loss rate, and a low D7, meaning poor information, adds a coordination overhead.

The third part is emergency extraction. The model assumes that some proportion of the population will have to be pulled out in an emergency at some point, and prices that in advance. The probability of this happening rises over time along a saturating curve: it starts near zero and approaches a ceiling. The daily rate driving that curve was fitted against historical cases and is 0.021 for Level 4, 0.010 for Level 3, 0.005 for Level 2 and 0.002 for Level 1. The Level 4 figure comes from averaging Mariupol in 2022 and Aleppo in 2016; the Level 3 figure from Mosul, Goma and the Central African Republic. The Level 2 and Level 1 figures have no historical anchor at all and are described as structurally plausible interpolations that require validation.

The curve is capped at 95 per cent for Level 4, 80 for Level 3, 60 for Level 2 and 30 for Level 1. Two modifiers then apply: a blocked corridor, meaning a low D3, raises the probability, and high urgency imposes a floor, 85 per cent when D6 is 5 and 60 per cent when D6 is 4. The floor exists because of Srebrenica, where the crisis unfolded over three days and no duration-based curve would have registered it.

The cost of extraction is anchored to the United Nations Humanitarian Air Service, which is the World Food Programme's air service flying humanitarian staff and light cargo into places commercial aviation does not serve. Its published operating cost was 2.08 US dollars per passenger kilometre in 2023. Ground extraction is priced at 30 per cent of that rate, which the code describes as an internal heuristic with no published source. Air

medical evacuation is priced at three times the rate, on the reasoning that a medical flight carries far fewer passengers. At Level 4 the whole extraction cost is multiplied by 2.5 for a helicopter premium, which the author records as unconfirmed, since UNHAS does not publish separate helicopter and fixed-wing rates.

Field medical treatment is costed separately. Injuries are estimated from a per-thousand daily rate that rises with the risk level: zero at Level 0, 0.1 at Level 1, 0.5 at Level 2, 2.0 at Level 3 and 8.0 at Level 4. The code states plainly that no public source reports in-field civilian injury rates as a daily incidence per thousand, so these are unvalidated. Each injury is costed at 800 US dollars, modified upward when the destination has poor medical capacity. The 800 figure sits within a documented peer-reviewed range: 211 US dollars per surgical case at a conflict-affected mission hospital in South Sudan in 2022, and roughly 500 to 650 dollars per case at two MSF surgical trauma centres around 2009, which inflate to roughly 780 and 1,010 dollars by 2026.

08

THE BREAK-EVEN ANALYSIS

This is the calculation the tool is built around. Two options are compared. Option A is to evacuate now and then provide assistance to whoever remains behind. Option B is not to evacuate and to assist the whole population in place. Option A carries a large one-time cost plus a small daily cost. Option B carries only a daily cost, but a larger one.

The break-even day is the one-time evacuation cost divided by the difference between the two daily costs. If evacuating costs 640,000 US dollars and staying costs 4,000 US dollars a day more than the post-evacuation arrangement, the break-even is 160 days. Before that day, staying is cheaper. After it, evacuation would have been cheaper.

It is a resource-planning number. The project's concept document is emphatic that it is not an answer to whether an evacuation is worth carrying out. The distinction is that the calculation compares the cost of one logistics operation against the cost of another logistics operation. It does not weigh either against lives. The decision to move civilians, the documentation states, is a matter of legal obligation and humanitarian judgment, not financial optimisation.

The tool also checks the transport mode against the dimension scores and warns when they are inconsistent. Five patterns trigger a warning: ground transport with a D4 of 4.0 or above, which indicates logistics breakdown; ground transport with a D1 of 4.5 or above, which indicates unacceptable kinetic exposure; walking with a population above 5,000, which is infeasible at scale; walking with a D2 of 3.5 or above, which means the population cannot walk; and air transport with a D3 of 2.0 or below, which means airspace authorization is missing. These are useful precisely because a cost model will happily produce a plausible-looking figure for an operation that could never be attempted.

09

ESTIMATING DEATHS: THE MORTALITY MODEL

Separately from cost, the tool estimates how many people would die and be injured if the population stayed. The author's framing is that this provides scale context for a planning decision, and both the README and the concept document state that the mortality model is indicative rather than predictive and that the financial estimates are substantially more reliable.

Each risk level has a baseline death rate expressed per 10,000 people per day. The current values are 0.777 at Level 0, 0.964 at Level 1, 3.625 at Level 2, 1.805 at Level 3 and 1.000 at Level 4. A reader will immediately notice that these are not in ascending order, which looks wrong and requires explanation. The explanation given in the code is that these are not standalone death rates but base rates that get multiplied by three further factors, and that Level 3 in the historical corpus is dominated by urban sieges and city fighting where a high proportion of the population is directly exposed, while Level 4 is dominated by large-enclave operations across much bigger populations where per-capita exposure is lower. The ordering was produced by the fitting procedure rather than assumed, and the author flags it explicitly as empirically validated but counterintuitive.

The confinement multiplier captures whether people are trapped. It is calculated as $5 - D3$, multiplied by $D4$, divided by 5. In plain words: when consent for movement is absent and logistics have collapsed at the same time, people cannot get out, and mortality rises sharply. The resulting score is converted into a multiplier in steps: half at the lowest, then 1, then 2, then 4, then 8 at the highest. An eightfold difference from a single factor is a very large lever, and its stepwise rather than smooth form is listed by the author as a known limitation. The anchors given are Aleppo and

Kosovo at a multiplier of 2, Mosul at 1, and the Kherson evacuation, which had an open corridor, at 0.5.

confinement score = $((5 - D3) \times D4) / 5$

The score is then mapped to a multiplier in steps: 0.5, 1, 2, 4, 8.

The displacement protection factor applies where part of the population has already left, since the remaining exposure is lower. The tool reduces the death rate by 60 per cent of the share that has departed. The 60 per cent coefficient, rather than 100, reflects that displaced people still face risk on the road, at checkpoints and from exposure. There is then an important refinement: in a siege, defined as a D3 of 2 or below combined with a D1 of 4 or above and a population of 500,000 or fewer, the coefficient is halved to 30 per cent. The reasoning is that in an encircled city, movement itself is lethal, because civilians pass checkpoints under fire and buses have been attacked. Displacement is still safer than staying, but only half as much safer as in an open corridor.

The geographic exposure factor recognises that not everyone in a conflict area is under fire at the same time. Four conflict shapes are recognised, with the fraction of the population treated as simultaneously exposed given in brackets: urban siege such as Mariupol or Aleppo (0.85), enclave such as Gaza (0.65), city conflict where a front line moves through a city, such as Mosul or Kherson (0.40), and regional dispersed conflict such as Sudan, the Central African Republic or the eastern Democratic Republic of the Congo (0.12). When the planner has not specified a shape, the tool infers one from D1 and the population size, on the logic that higher kinetic threat means more direct fire while a larger population means more dispersion. The population term uses a logarithm raised to the power 1.4, a refinement introduced specifically because the earlier formula did not fall away fast enough for continental-scale conflicts and overestimated Sudan by a wide margin.

Cumulative deaths accumulate linearly for the first 90 days and then decelerate along a square-root curve, on the reasoning that populations adapt and survivors relocate. This also marks the outer edge of the model's intended planning window. Injuries are set at four times deaths, following the ICRC planning ratio. The code notes that a peer-reviewed systematic review implies a ratio closer to 3.3 to 1, and that frontline-specific figures from Ukraine run far higher, and retains 4 to 1 as a mid-range planning estimate.

The infrastructure-denial multiplier applies only where there is primary source documentation that survival infrastructure was deliberately destroyed. It is calculated as 1 plus 0.4251 times D1 minus 3, times D4 minus 3, and only when D1 is at least 4.5 and D4 at least 4.0. It is switched on for four documented cases: Mariupol, Aleppo, Vukovar and Huambo, giving effective multipliers between roughly 1.6 and 2.3. The supporting documentation cited is a 2024 report on starvation as a method of warfare in Mariupol, the UN Commission of Inquiry on Syria for hospital bombing in Aleppo, ICTY proceedings for Vukovar, and Human Rights Watch and Amnesty reporting on Angola for Huambo. Critically, this multiplier is switched off by default for live scenarios. The code records why: an earlier version activated it automatically whenever the dimension thresholds were met, and calibration accuracy collapsed from 80 per cent to 20 per cent. It is a finding about deliberate atrocity, not a threshold that can be inferred from slider positions.

$$\text{infrastructure-denial multiplier} = 1 + (0.4251 \times (D1 - 3) \times (D4 - 3))$$

Applied only where D1 is at least 4.5 and D4 at least 4.0, and only with primary source documentation. Off by default for live scenarios.

10

HOW THE MODEL WAS FITTED, AND AGAINST WHAT

The repository contains thirty-one documented conflicts. Each record holds the population at risk, duration in days, recorded deaths, numbers displaced and remaining, the estimated vulnerable percentage with its own source and confidence rating, the distance to safety with its source, all seven dimension scores assigned retrospectively, corridor status and notes, key lessons, IHL issues, and a calibration block recording what the model predicted against what was recorded, across every model version.

Sixteen cases are used to fit the model: Mariupol 2022, Gaza 2023, Aleppo 2016, the Kherson evacuation 2022, Mosul 2016, eastern DRC and Goma 2024, the second battle of Grozny 1999, both battles of Fallujah 2004, the Jenin refugee camp 2002, the siege of Vukovar 1991, Gaza Cast Lead 2008, Gaza Protective Edge 2014, the siege of West Beirut 1982, the second Nagorno-Karabakh war 2020, and the siege of Huambo 1993.

Fifteen are excluded, and the reasons form a clear statement of where the model does not apply. Duration beyond the 90-day window excludes Sarajevo at 1,425 days, Kuito in Angola at over 540, Marawi at 154 and the final phase of the Sri Lankan Vanni offensive at 120. Dispersed regional conflict with no siege perimeter excludes the Central African Republic in 2014 and Sudan in 2023, the latter also on duration and on a population of six million. Deliberate massacre as the dominant mechanism of death, rather than attrition under fire, excludes Srebrenica 1995, Hue 1968, Manila 1945 and Bucha 2022. Open-corridor forced displacement, where civilians were made to move rather than prevented from moving, excludes Kosovo 1999. Recorded death counts too uncertain to fit against exclude Misrata 2011, where the contested range runs from 102 to 700.

Two exclusions are of a different character and the repository says so openly. Eastern Ghouta 2018 and Gaza Pillar of Defense 2012 are labelled challenge cases and were excluded, in the words of the data file, to preserve the version 7 metrics. They document a real structural limit: the Level 4 base rate cannot simultaneously fit a high-mortality urban siege such as Aleppo, with 31,000 deaths, and a large-enclave precision operation such as Cast Lead, with 965. The model undercounts one and overcounts the other by a wide margin. Retaining them in the corpus while removing them from the fit is a defensible research decision, and the fact that it is documented rather than concealed is to the project's credit. But a reader should understand that the headline accuracy figures are calculated on a set from which two known-difficult cases were removed.

The fitting procedure adjusts six numbers: the five base death rates and the infrastructure-denial coefficient. It searches for the values that minimise the mean squared error between the logarithm of predicted deaths and the logarithm of recorded deaths. Working in logarithms means the procedure fits proportional rather than absolute error, which is appropriate when the recorded death counts in the corpus run from 37 to 45,000. The search is constrained so that the rates for Levels 0, 1 and 2 rise in order; Levels 3 and 4 are deliberately left unconstrained relative to each other, which is how the counterintuitive ordering emerged.

On the sixteen fitted cases, the correlation between predicted and recorded deaths in logarithmic space gives an R-squared of 0.855, meaning the model captures most of the variation in order of magnitude. Seven of the sixteen cases, or 44 per cent, fall within a factor of two of the recorded figure. Those two statements sit oddly together and both should be kept in view: the model is good at ranking conflicts by scale of loss and much less good at predicting the number for any individual conflict. A battery of statistical

checks reported in the README covers normality of residuals, constant variance, rank correlation and multicollinearity, and reports Aleppo as the single most influential case.

One qualification about the cross-validation figure is worth making. The reported leave-one-out R-squared of 0.807 is not produced by refitting the whole model with each case removed in turn. It fits a straight line through the remaining points in logarithmic space and predicts the held-out point from that line. It therefore measures how stable the relationship between predicted and recorded deaths is across the corpus, which is useful, rather than how well fitted parameters generalise to unseen conflicts, which is what the term usually implies.

11

WHERE THE DATA COMES FROM

The Uppsala Conflict Data Program, based at Uppsala University in Sweden, is the principal academic provider of data on organised violence worldwide. Its Georeferenced Event Dataset, version 26.1, records individual lethal events down to the level of a single village on a single day, with separate counts for civilian deaths and for each party, and low, best and high estimates reflecting source uncertainty. ERCF uses it to check its historical cases against an independent record and to let a planner query recorded violence for a location and date range. The tool queries the live service where a token is available and falls back to a locally stored copy of the dataset otherwise.

ACAPS is an independent humanitarian analysis organisation. Together with the European Commission's Joint Research Centre it publishes the INFORM Severity Index, a monthly assessment scoring the severity of humanitarian crises across more than thirty indicators grouped into the impact of the crisis, the conditions of the people affected, and the complexity of the response. ERCF uses these scores to colour a world map by country-level severity and to convert a national severity score onto its own five-level scale. Two things should be noted. The repository works from a stored table of roughly eighty countries with a live overlay attempted on top, and the code records that the two relevant ACAPS endpoints were consistently timing out at the time of writing, which means in practice the stored table is usually what drives the map. Users should treat the map as an orientation aid rather than a live feed.

GeoNames provides city population figures so a planner can start from a real place rather than a guess. The project's own backlog flags the obvious limitation: these are pre-conflict populations and do not reflect displacement that has already occurred, so a user must adjust them manually.

Open-Meteo provides historical weather from the European reanalysis archive. ERCF uses fourteen days of temperature, precipitation, snowfall and wind to classify the operating conditions and to set two multipliers: one on fuel and transport, one on shelter. Deep winter raises fuel costs by 35 per cent and shelter costs by 120 per cent, the latter reflecting that cold-climate shelter requires more space per person and winterised units cost several times a standard tent. A heavy rainy season raises fuel and transport by 80 per cent, reflecting documented road impassability. If the planned date is in the future, the tool steps back a year at a time to find comparable historical conditions and labels the result a planning estimate. The weather data is tagged as validated; the multipliers derived from it are tagged as estimated.

A language model is used for one narrow purpose: generating a written country briefing that summarises the crisis, access conditions, likely exit routes, neighbouring safe areas, active humanitarian actors and the applicable legal framework, and proposes starting values for the seven dimensions. If the service is unavailable the tool falls back to fixed text built from the stored severity table. This is a convenience feature for a first draft of a scenario. Nothing in the cost or mortality calculation depends on it.

For eighteen conflict-affected countries the tool suggests a vulnerable population percentage. The figure is calculated as the percentage of children under five, plus the percentage of adults over sixty, plus half the percentage of people with disabilities. The halving is to avoid double-counting people who are both elderly and disabled, and to reflect that only some disabilities create evacuation-relevant constraints. The underlying figures come from UNICEF's State of the World's Children 2023 and the UN World Population Prospects 2022, with a uniform 15 per cent disability figure drawn from the WHO World Report on Disability. The results range from 21.5 per cent for Syria, Iraq and Libya to 36.5 per cent for Ukraine, where the driver is an unusually large elderly population. Two caveats matter: the child component counts only children under five rather than all minors, and the disability figure is a global average applied uniformly, whereas conflict-affected populations are known to have higher rates. The suggestion is advisory and the planner confirms or dismisses it.

Two supplementary transport models exist. Air evacuation prices fixed-wing movement at the same UNHAS rate of 2.08 US dollars per passenger kilometre and helicopter movement from a United Nations procurement contract for a Mi-8 airframe divided by an assumed number of guaranteed flight hours, returning a low, mid and high band rather than a single figure, with a deliberately conservative 50 per cent load factor. It plans fleet size across one, three, seven, fourteen and thirty day windows. Walking evacuation interpolates group speed between 8 kilometres per hour for healthy adults and 3 kilometres per hour for a fully vulnerable group, the latter taken from published gait-speed data for adults over 65, and assumes 8 walking hours a day rather than the 15 used in the academic migration model it draws on. It contains no attrition or mortality term at all, reports only the number of days of exposure, and states explicitly that the exposure does not feed back into the danger or urgency dimensions: that remains the analyst's judgment. Both models are marked as less validated than the ground transport model, and the backlog records that walking is not yet wired into the break-even calculation.

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READING THE RESULTS

The interface is a single web page with several sections. The Scenario Builder holds the seven sliders and the population, vulnerability, distance and terrain inputs, and updates every figure as the sliders move. A cost breakdown itemises the evacuation estimate line by line. A decision analysis panel shows the break-even chart with the two cost lines crossing. Historical Cases lists all thirty-one conflicts with badges marking which are out of scope and which are challenge cases. Compare on Radar overlays the current scenario's seven scores against any historical case as a seven-pointed shape, alongside a table comparing population, cost, cost per evacuee, duration and deaths. Map View colours the world by severity. A references section lists the data sources, and a methodology panel documents the statistical validation.

Experienced coordinators reason by analogy, saying that a situation resembles Mosul or resembles Mariupol. The radar overlay makes that comparison explicit and checkable, by showing exactly which of the seven dimensions match and which do not. That is a modest but genuine contribution: it turns an intuition into something a colleague can disagree with.

Three real operations are provided for cost benchmarking: the Kosovo Humanitarian Evacuation Programme in 1999, which moved 60,549 people at roughly 562 US dollars each; Libya in 2011 at roughly 984 dollars; and the evacuation of about 15,000 people from Lebanon to Canada in 2006 at several thousand dollars each. The README is careful to explain why these exceed the model's own figures by a wide margin: ERCF prices the immediate field operation only, at roughly 134 US dollars per person for a Level 2 ground movement over 50 kilometres, while real operations include international transport, reception facilities and weeks of hosting. A user comparing the tool's output to a real operation without reading that note would draw the wrong conclusion.

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GROUNDING IN INTERNATIONAL HUMANITARIAN LAW

Each risk level is tied to a stated legal reference. At Level 1 the tool cites the right of voluntary departure under Article 35 of the Fourth Geneva Convention. At Level 2 it cites the precautionary obligations in Articles 57 and 58 of Additional Protocol I, which require parties to take feasible precautions to spare civilians, including effective advance warning, and to keep civilians away from military objectives. At Level 3 it cites Article 49 of the Fourth Geneva Convention. At Level 4 it cites the same article's prohibition on forcible transfer and the corresponding provision of the Rome Statute of the International Criminal Court.

Article 49 is the central provision and deserves setting out. It prohibits individual or mass forcible transfers and deportations of protected persons from occupied territory, regardless of motive. It then creates a narrow exception: an Occupying Power may undertake total or partial evacuation of a given area if the security of the population or imperative military reasons so demand. Such evacuations must not move people outside the occupied territory unless it is materially impossible to avoid, and those evacuated must be returned home as soon as hostilities in the area have ceased. The ICRC commentary is emphatic that imperative means an absolute constraint leaving no other choice, not military convenience or advantage.

This matters for a cost tool because the exception in Article 49 turns on necessity, not on cost or convenience. A tool that produced a number and called it a recommendation would risk implying that an evacuation which is expensive is therefore not demanded, which inverts the legal test. ERCF's stated philosophy addresses this directly: labels are descriptive, not

imperative; a Level 4 classification describes a risk profile and does not order an action. The break-even day is presented as a resource-planning figure, not as a decision rule.

The framework also draws on Sphere standards throughout for its quantities. The Sphere Handbook is the humanitarian sector's most widely used set of minimum standards for response, developed collectively by humanitarian organisations, covering water and sanitation, food security, shelter and health. It is a standards document rather than a costing document, which is precisely the boundary ERCF has to cross: Sphere tells the model how many litres and how many square metres, but not what they cost.

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THE ETHICAL POSITION

A tool that computes the cost of evacuating people and the cost of not evacuating them invites an obvious objection. The repository addresses it in a notice displayed in the interface itself and repeated in the concept document: the tool estimates the operational cost of humanitarian evacuation logistics, in order to support planning and resource mobilisation, and does not place a monetary value on human life.

The structural safeguards behind that claim are worth naming, because they are design decisions rather than assertions. Cost and mortality are computed by separate functions and never combined into a single index; there is no cost-per-life-saved figure anywhere in the codebase. The mortality output is described in the documentation as contextual support only, not a target, threshold or measure of acceptable loss. The break-even calculation compares one logistics operation against another logistics operation, never against a count of deaths. And the concept document states explicitly that the tool is not intended to determine whether an evacuation is worth it or to rank the value of civilian lives.

The residual risk remains, and it is the risk that attaches to any costing tool in this domain: a figure produced for a funding appeal can be repurposed in an argument about whether to act. The separation in the software is real, but it depends on the people using the output to maintain the same separation.

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NOTABLE INCONSISTENCIES FOUND IN REVIEW

A plain-language review is of limited value if it only reports what the documentation claims. The following discrepancies were found by reading the code against the documentation. None is fatal, and several are the ordinary residue of a fast-moving research project, but a reader relying on the repository should know about them.

The acronym is used inconsistently. The README and both design documents give ERCF as the Evacuation Risk and Cost Framework. The prompt used for generating country briefings expands it as the Evacuation Risk Classification Framework. This report follows the README.

Comments have fallen out of step with the code. The explanation attached to the in-zone assistance calculation still lists access multipliers of 1.0, 1.5, 3.0, 5.0 and 8.0, while the values actually used are 1.0, 1.5, 2.0, 3.6 and 4.0. The same explanation gives a treatment cost of 1,200 US dollars per injury, while the code applies 800. In both cases the values in force are the more conservative and better-evidenced ones, so the effect is that the documentation overstates the model's outputs rather than understating them. A summary string displayed with the evacuation results still reports the Sphere emergency minimum of 15 litres of water per person per day although the calculation uses 20.

The stated date range of the corpus is wrong. The README describes the historical corpus as covering 1991 to 2024. The earliest case in the data file is the battle of Manila in 1945, and the corpus also includes Hue in 1968 and West Beirut in 1982. The correct range is 1945 to 2024.

One calibration script is not portable. It contains a file path pointing at a directory on the original author's own computer, which means it will not run on a fresh copy of the repository without editing that line. The main calibration script runs correctly. A comment in that script still refers to twenty in-scope cases, although it counts them dynamically and in practice uses sixteen.

One country code is wrong. The demographic dataset contains a country code entry for Mali that does not match that country's actual three-letter code, so a lookup by code would fail for Mali while a lookup by name succeeds.

MATURITY, AND WHAT REMAINS UNBUILT

The core of ERCF is genuinely built. The seven-dimension scoring, the evacuation cost model, the in-zone assistance model, the break-even analysis, the mortality model, the historical corpus, the calibration pipeline, the statistical validation, the transport warnings, the radar comparison and the map are all implemented and running. The calibration can be reproduced by anyone who clones the repository and runs a single command. This is not a mock-up.

What the project's own backlog lists as outstanding is instructive, and it divides into three groups. In the first are features that would improve the tool as it stands: making fuel, food and medical supply prices adjustable per country, showing the user when the external data was last updated, adding a structured flag for cases involving mass atrocity rather than leaving it in free text, and building a pipeline for adding new historical cases and re-running the fit. In the second are capabilities the tool does not currently have: the dimension scores are a static snapshot with no modelling of how a conflict escalates over time, only one evacuation route can be assessed at a time, walking evacuation is not integrated into the break-even calculation, and there is no export to a document format that a funding appeal could use directly. In the third are research questions: expert-panel validation of the seven weights, resolving the Level 4 structural limitation that the two challenge cases document, and putting confidence intervals around the cost estimates, which are currently single point figures.

That last item deserves emphasis. Every cost figure the tool produces is a single number built from parameters that the project itself classifies as validated, estimated or unvalidated. A total assembled from a validated water quantity, an uncited food price, an estimated tent price and an unvalidated access multiplier is not a number with uniform reliability, and at present the interface does not show a range around it.

A companion module, RICS, the Resettlement and Inclusion Capacity Simulator, sits in a subdirectory of the same repository. It applies the same cost and break-even machinery to a different question: the long-term cost of a protracted refugee camp against the upfront cost of local integration, with Dzaleka camp in Malawi as its worked case. Its architecture is built out, with roughly two thousand lines of Python and a working interface, and it introduces two design ideas worth noting: a requirement, enforced by the database itself rather than by policy, that no group smaller than twenty

people can be recorded, so that no individual can be identified; and a decision to propagate uncertainty into the cost estimate by widening each parameter into a band according to how well evidenced it is, rather than by simulation. But RICS is scaffolding at present. Several of its key parameters are unset placeholders, its readiness parameters are inherited wholesale from another tool and tagged unvalidated, and its own design document says that its central coefficient awaits calibration against real camp data. It should be read as a design proposal with working plumbing, not as a second finished tool.

A copy of a second Ethical Tech CoLab project sits in the repository as a reference. That tool, the Evacuation Readiness and Uncertainty Simulator, asks a different question: not what an evacuation costs, but how confident an assessment of readiness can be. It scores candidate destinations across seven factors, three of which are treated as gatekeepers, meaning that security, authority consent and host willingness each individually cap readiness at twenty per cent if blocked, no matter how good everything else looks. It runs five hundred simulations per pairing under a global uncertainty setting, in order to show that identical assessed readiness produces materially different predicted outcomes as confidence degrades. Its two epistemic distinctions are worth carrying into any reading of ERCF: that a confidently identified gap is better than an unreliably assessed one, and that Unknown and Unwilling must never be conflated, because an intelligence gap can be closed while a confirmed refusal cannot. The two projects share a documentation structure and an author, and they are the two parents of RICS, which ports the cost engine from ERCF and the readiness engine from the simulator. This report treats ERCF on its own terms; the simulator is not required to understand it.

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LIMITATIONS AND CAVEATS

Beyond the specific issues already described, the following limitations apply to the framework as a whole.

The weights are unvalidated. All seven are the author's modelled estimates. No expert panel has reviewed them.

The dimension scores are a snapshot. There is no modelling of escalation. A conflict that will look completely different in three weeks is scored as it looks today.

The planning window is 90 days. Beyond that, mortality accumulation is subject to a saturation adjustment and the model is outside the range of its own calibration data.

The mortality model fits scale, not individual cases. Fewer than half the fitted cases fall within a factor of two of their recorded death tolls. The model is useful for indicating whether a situation is in the hundreds, the thousands or the tens of thousands, and should not be used for anything finer.

The model has documented blind spots. It does not capture deliberate massacre or genocide, where death occurs in days rather than accumulating through attrition. It does not capture famine or the collapse of healthcare under blockade, which the code identifies as a major driver of mortality in Gaza that the seven dimensions do not represent. It underestimates open-corridor forced displacement, because it assumes a population trapped under fire rather than one compelled to move.

The most consequential cost multipliers are the least evidenced. The access multiplier that governs the entire in-zone assistance calculation is unconfirmed at its highest level, and the loss rates at Levels 2, 3 and 4, which reach fifty per cent, have no published equivalent that the author could locate.

Population figures are pre-conflict. City populations retrieved automatically do not account for displacement that has already taken place, and must be adjusted by hand.

The tool models one corridor. Real evacuation planning compares routes.

The historical corpus is small. Sixteen fitted cases is enough to sanity-check a framework, not enough to support statistical generalisation, and two known-difficult cases were held out of the fit.

Above all, this is a research prototype. It was produced by one researcher with academic supervision. Its own README states that it does not constitute operational advice and that all estimates require validation against country-specific intelligence and field assessment before any operational application.

INTENDED AUDIENCE AND USE

The concept document identifies five groups: humanitarian programme officers producing rapid cost estimates for funding proposals, emergency coordinators comparing scenarios before a crisis escalates, donors and fund managers benchmarking against documented operations, researchers studying civilian protection frameworks, and government civil protection agencies doing preparedness planning.

The most defensible use of the tool today is the one furthest from an active crisis: preparedness planning, funding appeal preparation, and teaching. Its value in those settings does not depend on the precision of any individual figure. It depends on forcing a planner to state, in numbers, what they are assuming about vehicle capacity, staff ratios, supply losses and access costs, and on making those assumptions available for a colleague to contest.

The tool is explicitly not a real-time conflict monitoring system, not a forecasting model, not a mass casualty prediction tool, and not a substitute for field assessment or expert judgment. The repository states each of these explicitly.

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CONCLUSION

ERCF's contribution is not a set of accurate numbers. It is the assembly of scattered, separately published operational standards into one calculation that can be run, inspected and argued with, together with an unusually disciplined habit of recording where each number came from and how far it can be trusted. The source code reads in places less like software than like a research notebook, documenting failed searches, revised assumptions, and figures that were lowered when the evidence would not support them. A model that reduced its own access multipliers from eight to four because no published source went above four is behaving the way a research instrument should.

The framework's structural insight is the separation it maintains between the danger a population faces and the practicality of moving it, carried through into the separation of cost from mortality and of description from recommendation. This keeps the legal question, which turns on necessity, from being quietly answered by an operational one, which turns on difficulty.

The prototype's weakest point is also clearly visible: the parameters carrying the most weight in the in-zone assistance calculation are the ones with the least published support behind them, and the mortality model resolves scale rather than magnitude. The author says so in the code, in the README and in the concept document. A reader should take the transparency as an invitation to check the work rather than as a reason to skip checking it.

What a tool of this kind can reasonably offer today is a structured first estimate, produced in an hour rather than a week, with every assumption on the surface where a humanitarian planner, a donor or a lawyer can see it and challenge it. That is a real contribution to a decision that is currently made largely from memory, and it is offered here without the claim that the numbers themselves are yet good enough to act on.

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